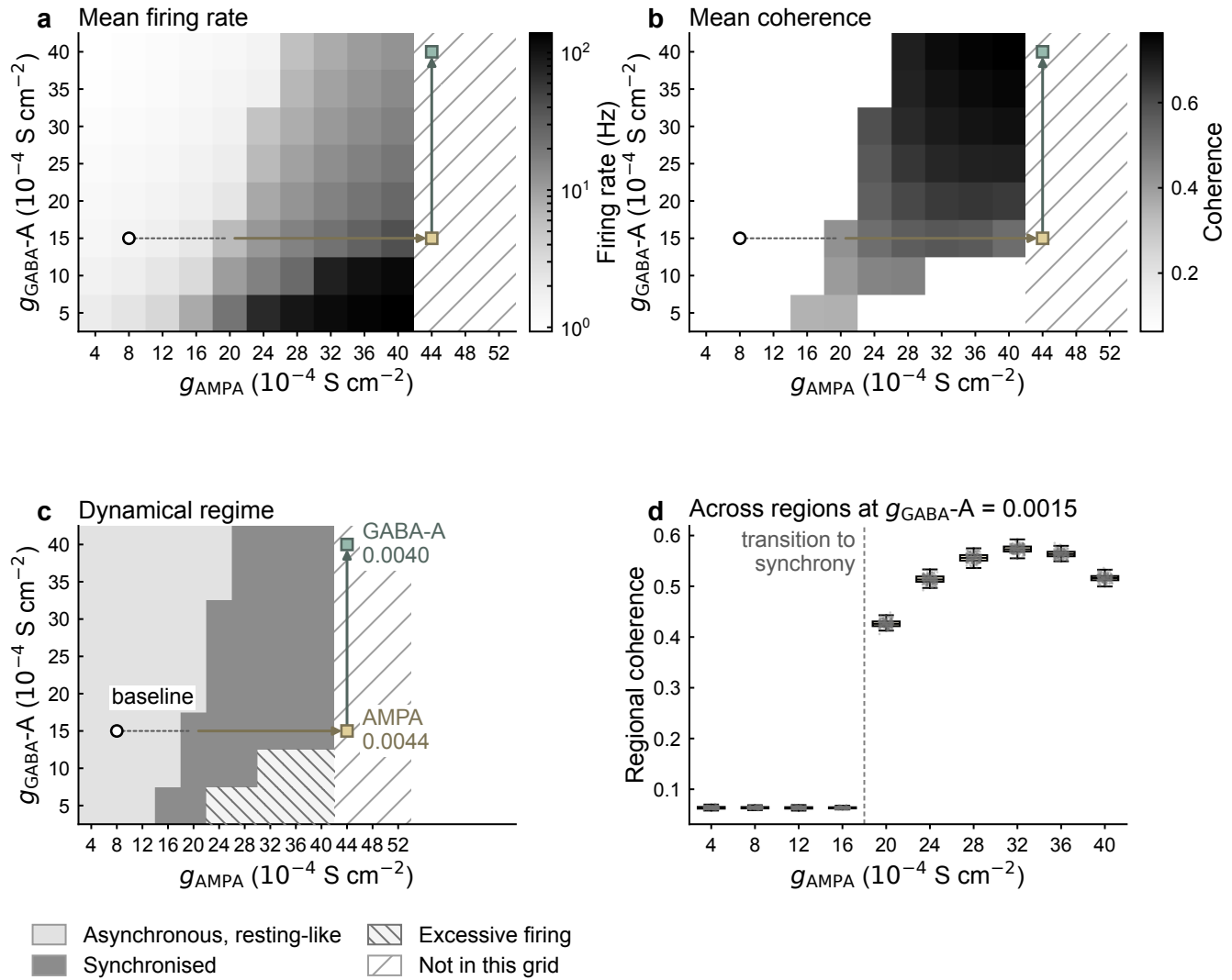




Supplementary Fig. S18 | Two-dimensional AMPA/GABA-A conductance sweep used to place the baseline and the perturbation endpoints. Each cell is one simulation of the digital twin brain at a fixed pair of global conductances ($n = 80$ simulations; unit of observation, one conductance pair; g_{AMPA} 0.0004-0.0040 S cm⁻² in steps of 0.0004 and $g_{\text{GABA-A}}$ 0.0005-0.0040 S cm⁻² in steps of 0.0005, external input 0). One run per cell, so no error indicator and no statistical test is shown; all four panels are descriptive maps of the model's dynamical regimes. Columns hatched in white were not simulated in this grid and are shown only so that the conductance settings used in the main analyses can be marked in place. **a**, Mean population firing rate (logarithmic colour scale, 0.93-138.81 Hz). **b**, Mean coherence across regions, which separates asynchronous from synchronised activity. **c**, Regime classification. Coherence is bimodal across the grid - 46 cells lie at the 0.0629-0.0654 floor and 34 cells at 0.348-0.763, with no cell in between - and the floor cells separate by rate just as cleanly, so the boundaries are read off these gaps rather than set by hand: asynchronous (up to 3.42 Hz, 38 cells), synchronised (5.28-39.86 Hz, 34 cells) and excessive firing (from 69.67 Hz, 8 cells), the last excluded from use. The open circle marks the baseline ($g_{\text{AMPA}} = 0.0008$, $g_{\text{GABA-A}} = 0.0015$ S cm⁻²; 1.54 Hz, coherence 0.0636), chosen in the low-firing, low-coherence resting-like regime; the arrows trace the protocol actually used, AMPA raised first to 0.0044 S cm⁻² (gold square) and GABA-A then raised to 0.0040 S cm⁻² on top of it (teal square), each value having been selected as the one producing the largest increase in the summed NP factor over the calibrated range (Methods). **d**, Coherence of each of the 217 regions along the AMPA axis at the baseline inhibition level ($g_{\text{GABA-A}} = 0.0015$ S cm⁻²), the line along which AMPA was swept: boxes show the median and interquartile range across regions with whiskers at 1.5x IQR and every region plotted ($n = 217$ regions per box; unit of observation, one region). Regions move into synchrony together rather than one at a time, and the spread across regions widens once the network synchronises (median interquartile range 0.0028 below the transition versus 0.010 above it); gold fill marks the synchronised settings. Descriptive; no test.